

hp - Finite Elements

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Let V_{hp} be a p -th order finite element sub-space of H^1 . By scaling and Bramble-Hilbert technique one obtains the best-approximation error estimate

$$\inf_{v_{hp} \in V_{hp}} \|u - v_{hp}\|_{H^1} \leq ch^{m-1} \|u\|_{H^m}$$

for $m \leq p + 1$. The constant c depends on the order p . If m is fixed, we do obtain reduction of the approximation error as we increase p . Next we develop methods to obtain so called p -version error estimates

$$\inf_{v_{hp} \in V_{hp}} \|u - v_{hp}\|_{H^1} \leq c \left(\frac{h}{p} \right)^{m-1} \|u\|_{H^m},$$

where c is independent of h and p . This estimate proves also convergence of the p -version finite element method: One may fix the mesh, and increase the order p .

A detailed analysis of local H^m norms allows an optimal balance of mesh-size h and polynomial order p . This hp -version leads to exponential convergence

$$\inf_{v_{hp} \in V_{hp}} \|u - v_{hp}\|_{H^1} \leq ce^{-N^\alpha},$$

where N is the number of unknowns.

We will prove the p -version estimate, but not the hp -result.

30.1. Legendre Polynomials

Orthogonal polynomials are important to construct stable basis functions for the p -FEM, as well as for error estimates.

Let Π_n denote the space of polynomials up to order n . We write π_n for a generic polynomial in Π_n , with a different value any time it appears.

Definition of Legendre polynomials via Rodrigues' formula:

$$P_n(x) := \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

It is a polynomial of degree n . The first few Legendre polynomials are

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{3}{2}x^2 - \frac{1}{2} \end{aligned}$$

P_n is even if n is even, and P_n is odd if n is odd. Since $(x^2 - 1)^n = x^{2n} - nx^{2n-2} + \pi_{2n-4}$ (with proper modification for small n) we have

$$P_n(x) = \frac{1}{2^n n!} \frac{(2n)!}{n!} x^n - \frac{n}{2^n n!} \frac{(2n-2)!}{(n-2)!} x^{n-2} + \pi_{n-4}$$

Lemma: Orthogonality

There holds

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1} \delta_{n,m}.$$

Proof: W.l.o.g. let $n \leq m$. Multiple integration by parts gives

$$\begin{aligned}
2^{n+m} n! m! \int_{-1}^1 P_n(x) P_m(x) dx &= \int_{-1}^1 \frac{d^n}{dx^n} (x^2 - 1)^n \frac{d^m}{dx^m} (x^2 - 1)^n dx \\
&= \int_{-1}^1 \frac{d^{n+1}}{dx^{n+1}} (x^2 - 1)^n \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^n + \left[\frac{d^n}{dx^n} (x^2 - 1)^n \right]_{-1}^1 \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^n \\
&= \dots \\
&= \int_{-1}^1 \frac{d^{n+m}}{dx^{n+m}} (x^2 - 1)^n (x^2 - 1)^m dx
\end{aligned}$$

For $n < m$, the first factor of the integrand vanishes, and we have orthogonality. For $n = m$ this equals

$$\begin{aligned}
2^{2n} (n!)^2 \|P_n\|_{L_2}^2 &= \int_{-1}^1 (2n)! (x^2 - 1)^n dx = (2n)! \int_{-1}^1 (x - 1)^n (x + 1)^n dx \\
&= - (2n)! \int_{-1}^1 \frac{n}{n+1} (x - 1)^{n+1} (x + 1)^{n-1} dx \\
&= (2n)! \int_{-1}^1 \frac{n(n-1)}{(n+1)(n+2)} (x - 1)^{n+2} (x + 1)^{n-2} dx = \dots \\
&= (2n)! \frac{n!}{2n(2n-1) \dots (n+1)} \int_{-1}^1 (x - 1)^{2n} dx = (n!)^2 \frac{1}{2n+1} 2^{2n+1},
\end{aligned}$$

which proves the scaling. $\square$

Next we prove the 3-term recurrency, which can be used for efficient evaluation.

Lemma: Three term recurrence:

There holds

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x). \quad (30.1)$$

Proof: Set $r(x) = (n+1)P_{n+1}(x) - (2n+1)xP_n(x) + nP_{n-1}(x)$. Using $\text{\ref{equ_leadingcoef}}$, we see that the leading coefficients cancel, and thus $r \in \Pi_{n-2}$. From Lemma~\ref{lemma_ortho} we get for any $q \in \Pi_{n-2}$

$$\int_{-1}^1 r(x)q(x) dx = (n+1) \int_{-1}^1 P_{n+1} q - (2n+1) \int_{-1}^1 \underbrace{P_n xq}_{\in \Pi_{n-1}} + n \int_{-1}^1 P_{n-1} q = 0,$$

and thus $r = 0$. $\square$

Lemma: Sturm-Liouville

Legendre polynomials satisfy the Sturm-Liouville differential equation

$$\frac{d}{dx} \left[(x^2 - 1) \frac{d}{dx} P_n(x) \right] = n(n+1) P_n(x)$$

Proof: Both sides are in Π_n . We compare leading coefficients, for this set $P_n = a_n x^n + \pi_{n-2}$ (with $a_n = \frac{1}{2^n n!} \frac{(2n)!}{n!}$).

$$\begin{aligned} lhs &= \frac{d}{dx} \left[(x^2 - 1) \frac{d}{dx} (a_n x^n + \pi_{n-2}) \right] \\ &= \frac{d}{dx} [(x^2 - 1)(a_n n x^{n-1} + \pi_{n-3})] \\ &= \frac{d}{dx} [a_n n x^{n+1} + \pi_{n-1}] \\ &= n(n+1) a_n x^n + \pi_{n-2}, \end{aligned}$$

and we get the same leading coefficient for rhs. Furthermore, for $q \in \Pi_{n-1}$ there holds

$$\begin{aligned} \int_{-1}^1 lhs q &= - \int_{-1}^1 (x^2 - 1) P_n' q' dx + \underbrace{[(x^2 - 1) P_n' q]_{-1}^1}_{=0} \\ &= \int_{-1}^1 \underbrace{P_n ((x^2 - 1) q')'}_{\in \Pi_{n-1}} dx - [P_n (x^2 - 1) q']_{-1}^1 = 0, \end{aligned}$$

and the same for the rhs. Thus the identity is proven. $\backslash\text{end}\{\text{proof}\}$

Lemma~\ref{lemma_sturmliouville} implies that the Legendre polynomials are also orthogonal w.r.t. $(u', v')_{L_2, 1-x^2}$, i.e. $\int_{-1}^1 (1-x^2) P_n' P_m' = n(n+1) \|P_n\|_{L_2}^2 \delta_{n,m}$

30.2. Error estimate of the L_2 projection

Since polynomials are dense in $L_2(-1, 1)$, we get

$$u = \sum_{n=0}^{\infty} a_n P_n$$

with the generalized Fourier coefficients

$$a_n = \frac{(u, P_n)_{L_2}}{\|P_n\|_{L_2}^2},$$

and

$$\|u\|_{L_2}^2 = \sum_{n=0}^{\infty} a_n^2 \|P_n\|^2.$$

Let $P_{L_2}^{\Pi_p}$ denote the L_2 -projection onto Π_p . There holds

$$P_{L_2}^{\Pi_p} u = \sum_{n=0}^p a_n P_n$$

The projection error is

$$\|u - P_{L_2}^{\Pi_p} u\|_{L_2}^2 = \sum_{n=p+1}^{\infty} a_n^2 \|P_n\|^2$$

Lemma Projection error:

The L_2 -projection error satisfies

$$\|u - P_{L_2}^{\Pi_p} u\|_{L_2(-1,1)} \leq \frac{1}{\sqrt{(p+1)(p+2)}} |u|_{H^1(-1,1)} \quad (30.2)$$

Proof: Since P_n are orthogonal also w.r.t. $(u', v')_{L_2, 1-x^2}$, there holds

$$\|u'\|_{L_2, 1-x^2}^2 = \sum_{n \in \mathbb{N}} a_n^2 \|P'_n\|_{1-x^2}^2,$$

provided that u is in H^1 . The projection error satisfies

$$\begin{aligned} \|u - P_{L_2}^{\Pi_p} u\|_{L_2}^2 &= \sum_{n > p} a_n^2 \|P_n\|_{L_2}^2 = \sum_{n > p} a_n^2 \frac{1}{n(n+1)} \|P'_n\|_{1-x^2}^2 \\ &\leq \frac{1}{(p+1)(p+2)} \sum_{n > p} a_n^2 \|P'_n\|_{1-x^2}^2 \leq \frac{1}{(p+1)(p+2)} \sum_{n \in \mathbb{N}} a_n^2 \|P'_n\|_{1-x^2}^2 \\ &= \frac{1}{(p+1)(p+2)} \|u'\|_{1-x^2}^2 \end{aligned}$$

Finally, the result follows from

$$\int_{-1}^1 (1-x^2)(u')^2 dx \leq \int_{-1}^1 (u')^2 dx.$$

□

Similar as in Lemma~\ref{lemma_sturmliouville} one shows also

$$\frac{d^m}{dx^m} \left[(x^2 - 1)^m \frac{d^m}{dx^m} P_n(x) \right] = (n+m)(n+m-1) \dots (n-m+1) P_n(x)$$

for $m \leq n$, and, as in Lemma~\ref{lemma_l2est}

$$\|u - P_{L_2}^{\Pi_p} u\|_{L_2} \leq \sqrt{\frac{(p-m+1)!}{(p+m+1)!}} |u|_{H^m}.$$

30.3. Orthogonal polynomials on triangles

Orthogonal polynomials on tensor product elements are simply constructed by tensorization. Orthogonal polynomials on simplicial elements are more advanced. They are

based on Jacobi polynomials:

For $\alpha, \beta > -1$, Jacobi polynomials are defined by

$$P_n^{(\alpha, \beta)}(x) := \frac{(-1)^n}{2^n n!} \frac{1}{w(x)} \frac{d^n}{dx^n} (w(x)(1-x^2)^n)$$

with the weight function

$$w(x) = (1-x)^\alpha (1+x)^\beta.$$

Jacobi polynomials are orthogonal w.r.t. the weighted inner product

$$\int_{-1}^1 w(x) P_n^{(\alpha, \beta)}(x) P_m^{(\alpha, \beta)}(x) dx = \delta_{n,m} \frac{2^{\alpha+\beta+1}}{2n + \alpha + \beta + 1} \frac{\Gamma(n + \alpha + 1) \Gamma(n + \beta + 1)}{n! \Gamma(n + \alpha + \beta + 1)}.$$

Note that $P_n^{(0,0)} = P_n$.

Define the unit-triangle T with vertices $(-1, 0)$, $(1, 0)$ and $(0, 1)$.

Lemma: Dubiner basis

The functions

$$\varphi_{i,j}(x, y) := P_i\left(\frac{x}{1-y}\right) (1-y)^i P_j^{(2i+1, 0)}(2y-1) \quad i+j \leq p$$

form an $L_2(T)$ -orthogonal basis for $\Pi_p(T)$.

Proof: Note that $\varphi_{i,j} \in \Pi_{i+j}$. Substitution $\xi = \frac{x}{1-y}$ leads to

$$\begin{aligned}
\int_T \varphi_{ij}(x, y) \varphi_{kl}(x, y) d(x, y) &= \int_0^1 \int_{-1+y}^{1-y} P_i\left(\frac{x}{1-y}\right) (1-y)^i P_j^{(2i+1,0)}(2y-1) P_k\left(\frac{x}{1-y}\right) (1-y)^k \\
&= \int_0^1 \int_{-1}^1 P_i(\xi) P_k(\xi) (1-y)^{i+k+1} P_j^{(2i+1,0)}(2y-1) P_l^{(2k+1,0)}(2y-1) \\
&= \delta_{i,k} \|P_i\|_{L_2}^2 \int_0^1 (1-y)^{2i+1} P_j^{(2i+1,0)}(2y-1) P_l^{(2i+1,0)}(2y-1) \\
&= C_{ij} \delta_{i,k} \delta_{j,l}
\end{aligned}$$

□

30.4. Projection based interpolation

By means of the orthogonal polynomials one shows approximation error estimates of the form $\inf_{q \in \Pi_p(T)} \|u - q\|_{H^k(T)} \leq c p^{k-m} |u|_{H^m(T)}$ $m \geq k$, with $c \neq c(p)$, easily in 1D and tensor product elements, and also soon in d -dimensional simplices [Braess+Schwab: Approximation on simplices with respect to weighted Sobolev norms, J. Approximation Theory 103, 329-337 (2000)].

But, an interpolation operator to an H^1 -conforming finite element space has to satisfy continuity constraints across element boundaries. We show that we get the same rate of convergence under these constraints.

30.4.1. The 1D case

Let $I = (-1, 1)$. We define the operator $I_p : H^1(I) \rightarrow \Pi_p$ such that

$$\begin{aligned}
I_p u(x) &= u(x) & x &\in \{-1, 1\} \\
\int_I (I_p u)' q' &= \int_I u' q' & \forall q &\in \Pi_{p,0}(I),
\end{aligned}$$

where $\Pi_{p,0}(D) := \{q \in \Pi_p(D) : q = 0 \text{ on } \partial D\}$. This procedure is exactly a p -version Galerkin-method for the Dirichlet problem. Since boundary values are preserved, the interpolation operator produces a globally continuous function. The operator I_p is a kind of mixture of interpolation and projection, thus the term *projection based interpolation* introduced by Demkowicz has been established.

Lemma: **Commuting diagram** There holds

$$\Pi_{L_2}^{\Pi_{p-1}} u' = (I_p u)'$$

Proof: The range of both sides is Π_{p-1} . We have to show that $(I_p u)'$ is indeed the L_2 -projection of u' , i.e.

$$\int_I (I_p u)' q = \int_I u' q \quad \forall q \in \Pi_{p-1}.$$

This holds since $\{q' : q \in \Pi_{p,0}\} = \{q \in \Pi_{p-1} : \int q = 0\}$ and $(\text{ref}\{\text{equ_projbased2}\})$, and

$$\int_{-1}^1 (I_p u)' 1 = (I_p u)(1) - (I_p u)(-1) = u(1) - u(-1) = \int_{-1}^1 u' 1.$$

□

The H^1 -error estimate follows directly from the commuting diagram property:

$$|u - I_p u|_{H^1(I)} = \|u' - (I_p u)'\|_{L_2} = \|u' - P_{L_2}^{\Pi_{p-1}} u'\|_{L_2} \leq \frac{c}{p^{m-1}} |u'|_{H^{m-1}}.$$

By the Aubin-Nitsche technique one obtains an extra p for the L_2 -error:

$$\|u - I_p u\|_{L_2(I)} \preceq \frac{1}{p} |u - I_p u|_{H^1(I)} \leq \frac{1}{p^m} |u|_{H^m(I)}$$

One also gets for $q \in \Pi_p$

$$|u - I_p u|_{H^1} = |u - q - I_p(u - q)|_1 \leq \|Id - I_p\|_{H^1 \rightarrow H^1} |u - q|_{H^1}$$

30.4.2. Projection based interpolation on triangles

We define the operator $I_p : H^2(T) \rightarrow \Pi_p(T)$ as follows:

$$\begin{aligned}
I_p u(x) &= u(x) \quad \forall \text{ vertices } x \\
\int_E \partial_\tau(I_p u) \partial_\tau q &= \int_E \partial_\tau u \partial_\tau q \quad \forall \text{ edges } E, \forall q \in \Pi_{p,0}(E) \\
\int_T \nabla(I_p u) \nabla q &= \int_T \nabla u \nabla q \quad \forall q \in \Pi_{p,0}(T)
\end{aligned} \tag{30.3}$$

Note that $I_p u$ on the edge E depends only on $u|_E$, and thus the interpolant is continuous across neighbouring elements.

Lemma: Let $v \in C(\partial T)$ such that $v|_E \in \Pi_p(E)$. Then there exists an extension $\tilde{v} \in \Pi_p(T)$ such that $\tilde{v}|_{\partial T} = v$ and

$$|\tilde{v}|_{H^1} \leq c |v|_{H^{1/2}(\partial T)},$$

where c is independent of p .

Major steps have been shown in exercises 5.2 and 6.6. Note that the minimal-norm extension $\tilde{v}$ is the solution of the Dirichlet problem, i.e. $\int_T \nabla \tilde{v} \nabla w = 0 \quad \forall w \in \Pi_{p,0}$

Theorem: error estimate

There holds

$$|u - I_p u|_{H^1} \preceq \inf_{q \in \Pi_p} |u - q|_{H^1(T)} + \sum_{E \subset \partial T} \frac{1}{\sqrt{p}} \inf_{q \in \Pi_p(E)} |u - q|_{H^1(E)} \preceq \frac{1}{p^{m-1}} |u|_{H^m}$$

for $u \in H^m, m \geq 2$.

Proof: Let u_p be the $|\cdot|_{H^1}$ best approximation to u , i.e.

$$\int_T \nabla u_p \nabla v = \int_T \nabla u \nabla v \quad \forall v \in \Pi_p,$$

and, for uniqueness, mean values are preserved: $\int_T u_p = \int_T u$. There holds

$$|u - u_p|_{H^1} \leq \frac{c}{p^{m-1}} |u|_{H^m}.$$

We apply the triangle inequality:

$$|u - I_p u|_{H^1} \leq |u - u_p|_{H^1} + |u_p - I_p u|_{H^1}$$

Since

$$\int_T \nabla u_p \nabla v = \int_T \nabla u \nabla v = \int_T \nabla I_p u \nabla v \quad \forall v \in \Pi_{p,0}(T),$$

we have that

$$u_p - I_p u \perp_{H^1} \Pi_{p,0},$$

i.e. $u_p - I_p u$ is solution of the Dirichlet problem with boundary values $(u_p - I_p u)|_{\partial T}$.

Lemma~\ref{lemma_polext} implies that

$$|u_p - I_p u|_{H^1(T)} \preceq |u_p - I_p u|_{H^{1/2}(\partial T)}$$

We insert an u on the boundary to obtain

$$\begin{aligned} |u - I_p u|_{H^1} &\preceq |u - u_p|_{H^1(T)} + |u_p - I_p u|_{H^{1/2}(\partial T)} \\ &\leq |u - u_p|_{H^1(T)} + |u_p - u|_{H^{1/2}(\partial T)} + |u - I_p u|_{H^{1/2}(\partial T)} \\ &\leq |u - u_p|_{H^1(T)} + \|u - I_p u\|_{L_2(\partial T)}^{1/2} |u - I_p u|_{H^1(\partial T)}^{1/2}. \end{aligned}$$

In the last step we used that $H^{1/2}(\partial T) = [L_2, H^1]_{1/2}$ (i.e. the interpolation space). Next, we observe that I_p restricted to one edge E is exactly the 1D operator. Using Aubin-Nitsche we get

$$\begin{aligned} |u - I_p u|_{H^1} &\preceq |u - u_p|_{H^1(T)} + p^{-1/2} \|u - I_p u\|_{H^1(\partial T)} \\ &\preceq |u - u_p|_{H^1(T)} + \sum_E p^{1-m} |u|_{H^{m-1/2}(E)} \\ &\preceq p^{1-m} |u|_{H^m(T)} \end{aligned}$$

In the last step we used the trace theorem. $\square$